

Laboratório 03:

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2026-08-31

```
library(readr)
library(dplyr)
library(tidyr)
library(purrr)
library(leaflet)
```

Importação dos Dados

```
airlines <- read_csv("airlines.csv", show_col_types = FALSE)

airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE),
  show_col_types = FALSE
)
```

Processamento do Arquivo de Voos

```
processar_bloco <- function(bloco, pos) {
  bloco %>%
    select(DESTINATION_AIRPORT, ARRIVAL_DELAY) %>%
    filter(!grepl("^1", DESTINATION_AIRPORT)) %>%
    drop_na() %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atraso = sum(ARRIVAL_DELAY),
      n_voos = n(),
```

```

    .groups = "drop"
  )
}

estatisticas_blocos <- read_csv_chunked(
  "flights.csv",
  callback = DataFrameCallback$new(processar_bloco),
  chunk_size = 1000000,
  progress = FALSE,
  show_col_types = FALSE
)

flights <- estatisticas_blocos %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    soma_atraso = sum(soma_atraso),
    n_voos = sum(n_voos),
    MEDIA_ATRASO = soma_atraso / n_voos,
    .groups = "drop"
  ) %>%
  select(DESTINATION_AIRPORT, MEDIA_ATRASO)

```

Operações do Tipo Join

```

aeroportos_ausentes <- flights %>%
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

print(aeroportos_ausentes)

```

```

# A tibble: 0 x 2
# i 2 variables: DESTINATION_AIRPORT <chr>, MEDIA_ATRASO <dbl>

```

```

flights <- flights %>%
  inner_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

```

Aeroportos por Estado

```

aeroportos_por_estado <- airports %>%
  count(STATE, name = "QTD_AEROPORTOS") %>%
  arrange(desc(QTD_AEROPORTOS))

print(aeroportos_por_estado)

```

```

# A tibble: 54 x 2
  STATE QTD_AEROPORTOS
  <chr>      <int>
1 TX             24
2 CA             22
3 AK             19
4 FL             17
5 MI             15
6 NY             14
7 CO             10
8 MN              8
9 MT              8
10 NC             8
# i 44 more rows

```

Mapa dos Atrasos Médios

```

this <- leaflet(flights) %>%
  addTiles() %>%
  addCircleMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      "<b>Aeroporto:</b> ", DESTINATION_AIRPORT, "<br>",
      "<b>Cidade/UF:</b> ", CITY, "-", STATE, "<br>",
      "<b>Atraso Médio:</b> ", round(MEDIA_ATRASO, 2), " min"
    ),
    radius = ~pmax(3, pmin(MEDIA_ATRASO, 15)),
    stroke = FALSE,
    fillOpacity = 0.6
  )

this

```


Data e Hora da Compilação

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 14:31:34 -03"
```